



UNITED INTERNATIONAL UNIVERSITY (UIU)
DIGITAL SIGNAL PROCESSING LABORATORY
COURSE CODE EEE 3310

EXPERIMENT NO. 2

DISCRETE TIME SIGNALS

Prerequisite

Before starting this section, student should know:

1. A fundamental knowledge on MATLAB
2. A theoretical knowledge on discrete time signals

Lab Outcome

After Finishing this lab student should be able to:

1. Describe discrete-time signals mathematically and generate, operate and plot discrete-time signals using MATLAB
2. Understand different operations on DT signals

2.1 Generation of DT signals using MATLAB

Use custom function in this section:

Table 2.1

	Name	Mathematical definition	MATLAB function
1 .	Unit sample function	$\delta(n) = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$	<code>function [x n]= delta(n0,n1,n2) n=[n1:n2]; x=[n-n0]==0;</code>
2 .	Unit step function	$u(n) = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$	<code>function [x n]= u(n0,n1,n2) n=[n1:n2]; x=[n-n0]>=0;</code>
3 .	Real valued exponential sequence	$x(n) = a^n, \forall n; a \in \mathbb{R}$	<code>function [x n]= realx(a,n1,n2) n=[n1:n2]; x=a.^n;</code>

Example 2.1:

Plot the signal shown in Figure 1, using MATLAB code and function from table 2.1, where the delta signal is at $n=4$:

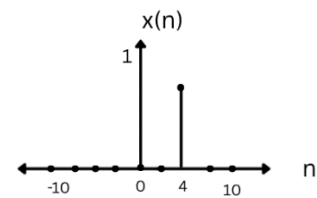


Figure 1: Delta Signal at $n=4$

Generating and plotting a delta signal using MATLAB code

```
clear all; close all;
```

```
% delta function
```

```
n0= 4;
```

```
n1= -10;
```

```
n2= 10;
```

```
[x n]= delta(n0,n1,n2);
```

```
figure
```

```
stem(n,x)
```

```
axis([-11,11,-1,2]); ylabel('x(n)'); xlabel('Number of Samples (n)');
```

```
title('Delta Signal'); fontsize(16, "points");
```

Output of Example 2.1

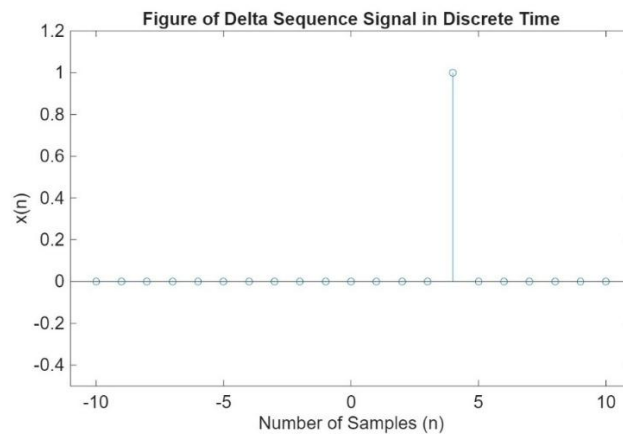


Figure 2: Delta Signal Plot using MATLAB

Example 2.2:

Plot the signal in Figure 2, using MATLAB code and function from table 2.1, where the delta signal is at $n = -5$, $n = 0$ and $n = 5$:

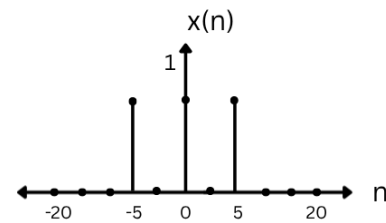


Figure 3: Delta Signal Sequence

Generating and plotting a delta signal using MATLAB code

```
clear all; close all;
%% delta sequence generation and plot
n01= -5;
n02= 0;
n03= 5;

n1= -20;
n2= 20;

[x1 n]= delta(n01,n1,n2);
[x2 n]=delta(n02,n1,n2);
[x3 n]=delta(n03,n1,n2);
x= x1+x2+x3;

figure
stem(n,x)
axis([-21,21,-0.5,1.2]); ylabel('x(n)'); xlabel('Number of Samples (n)');
title('Figure of Delta Sequence Signal in Discrete Time'); fontsize(16,
"points");
```

Output of Example 2.2

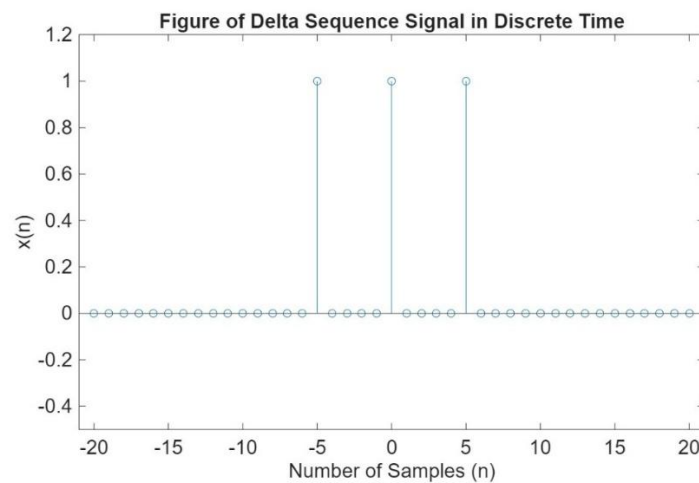


Figure 4: Delta Sequence plot using MATLAB

Example 2.3:

Plot the signal in figure 5, using MATLAB code and function from table 2.1, where the step signal starts from $n=5$ and ends to $n=10$:

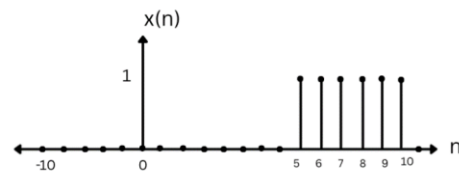


Figure 5: Unit Step Signal at $n=5$

Generating and plotting a step signal using MATLAB code

```
clear all; close all;
%% Step Signal generation and plot
n0=5;
n1=-10;
n2=10;
```

```
[x n]= unit_step(n0,n1,n2);
```

```
figure
stem(n,x)
axis([-11,11,-0.5,1.2]); ylabel('x(n)'); xlabel('Number of Samples (n)');
title('Figure of Step Signal in Discrete Time'); fontsize(16, "points");
```

Output of Example 2.3

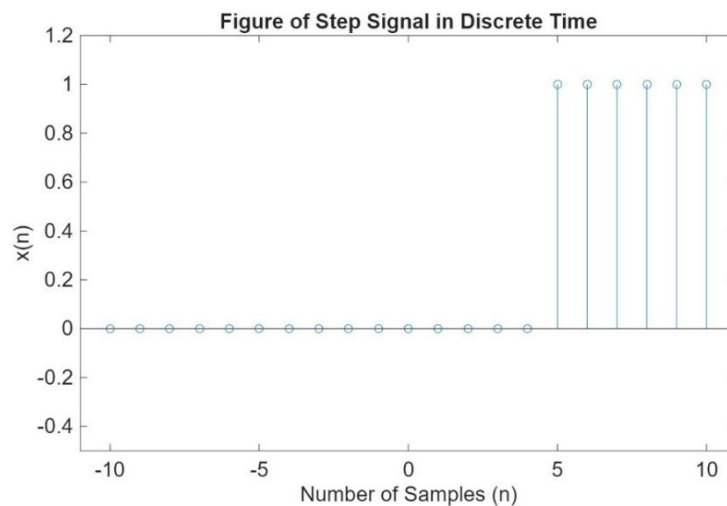


Figure 6 : Unit Step Signal Plot using MATLAB

Example 2.4:

Plot the following signal using MATLAB code where the step signal starts from $n=-2$ and ends at $n=2$ with the range from -10 to 10:

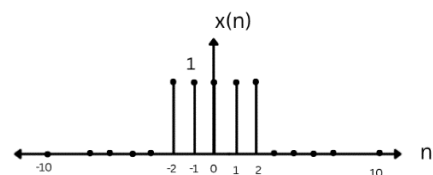


Figure 7: Window using Unit Step Signal

Generating and plotting a step signal using MATLAB code

```
clear all; close all;
%% Step Signal generation and plot
n01=-2;
n02=3;
n1=-10;
n2=10;

[x1 n]= unit_step(n01,n1,n2);
[x2 n]= unit_step(n02,n1,n2);

x= x1-x2;

figure
stem(n,x)
axis([-11,11,-0.5,1.2]); ylabel('x(n)'); xlabel('Number of Samples (n)');
title('Figure of Step Signal in Discrete Time'); fontsize(16, "points");
```

Output of Example 2.4

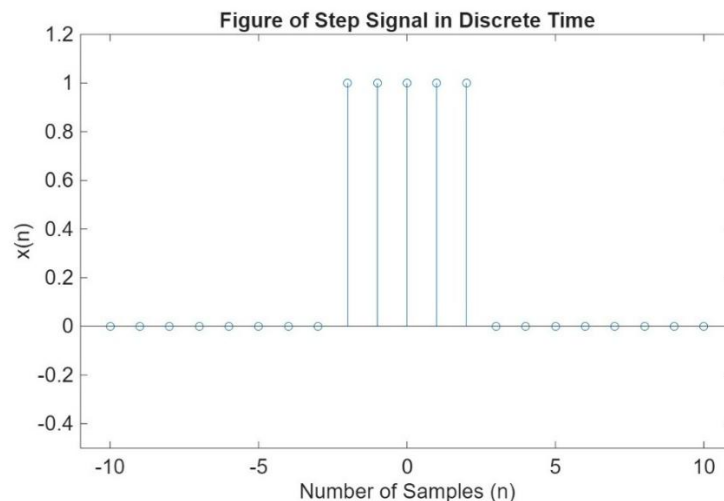


Figure 8: Plot of unit step function using MATLAB

Example 2.5:

Plot the following signal using MATLAB code and function from table 2.1, where the exponential signal is in the range from -10 to 10 with exponent $a=0.5$.

Generating and plotting an exponential signal using MATLAB code

```
clear all; close all;
%% Exponential function
a= 0.5;
n1= -10;
n2= 10;

[x n]= real_exp(a,n1,n2);
```

```
figure
stem(n,x)
ylabel('x(n)'); xlabel('Number of Samples (n)');
title('Figure of Exponential Signal in Discrete Time'); fontsize(14,
"points");
```

Output of Example 2.5

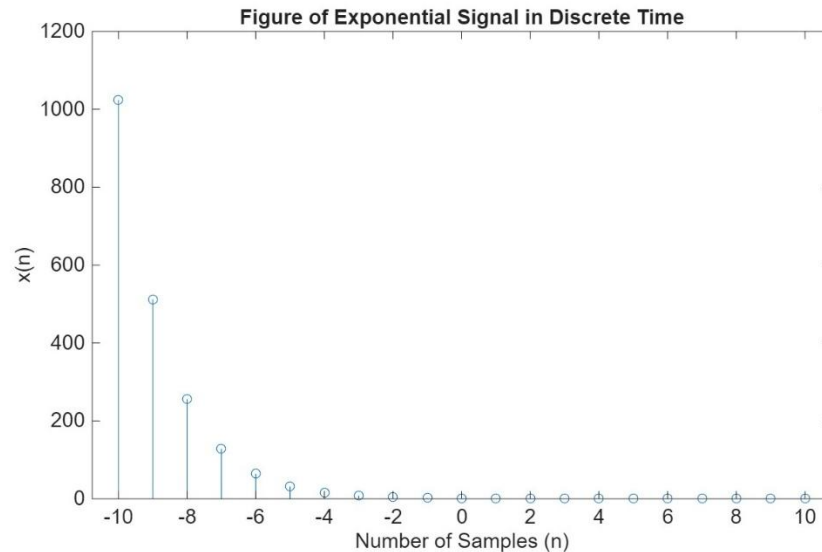


Figure 9 : Plot of Real Exponential signal using MATLAB

2.2 Windowing a signal using MATLAB

Windowing is the process of multiplying a time-domain signal by a window function, which is created using “Unit Step” function shown in example 2.4.

Example 2.6:

Perform windowing of the exponential signal created in example 2.5, where the window will only keep the positive time samples of the exponential signal starting from zero.

Windowing using unit step function with MATLAB code

```
clear all; close all;
%% %% Exponential function
a= 0.5;
n1= -10;
n2= 10;

[x n]= real_exp(a,n1,n2);
%% Windowing function
n01= 0;

[x1 n]= unit_step(n01,n1,n2);
y= x.*x1;

figure
stem(n,y, LineWidth=2)
```

```

hold on
stem(n,x1,'r')
axis([-11,11, -0.2, 1.2]);
xlabel('Number of Samples (n)'); legend('Window Output', 'Window')
title('Windowed Exponential Function'); fontsize(12, "points");

```

Output of Example 2.6

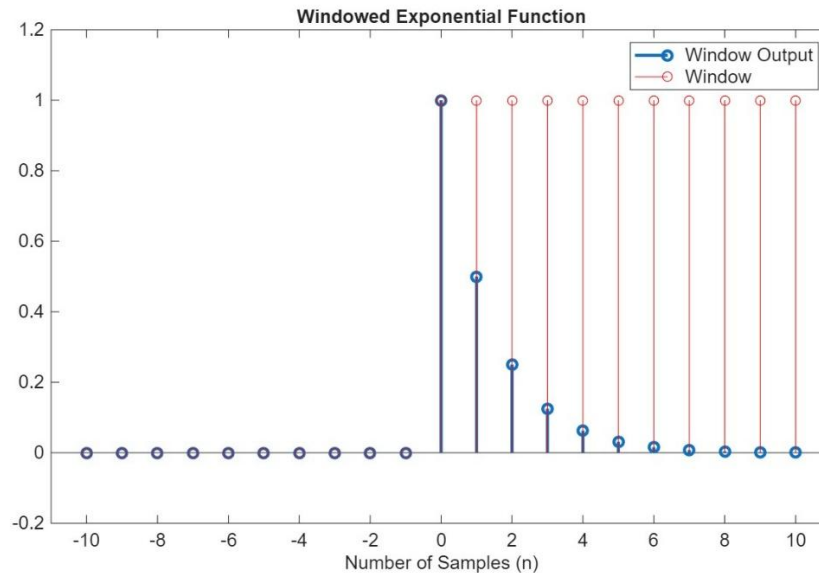


Figure 10 : Windowing using unit step function in MATLAB

2.3 Operations on DT signals Using MATLAB

Use custom function in this section:

Table 1.2

	Name	Mathematical definition	MATLAB function
1	Signal addition	$\{x_1(n)\} + \{x_2(n)\} = \{x_1(n) + x_2(n)\}$	<pre> function [y n]=sigadd(x1,n1,x2,n2) n=min(min(n1),min(n2)):max(max(n1),max(n2)); y1=zeros(1,length(n));y2=y1; y1(find((n>=min(n1)) & (n<=max(n1))==1))=x1; y2(find((n>=min(n2)) & (n<=max(n2))==1))=x2; y=y1+y2; end </pre>
2	Signal multiplication	$\{x_1(n)\} \cdot \{x_2(n)\} = \{x_1(n) \cdot x_2(n)\}$	<pre> function [y n]=sigmult(x1,n1,x2,n2) n=min(min(n1),min(n2)):max(max(n1),max(n2)); y1=zeros(1,length(n));y2=y1; y1(find((n>=min(n1)) & (n<=max(n1))==1))=x1; y2(find((n>=min(n2)) & (n<=max(n2))==1))=x2; y=y1.*y2; end </pre>
3	Signal Shifting	$y(n) = \{x(n-k)\}$	<pre> function [y n]=sigshift(x,m,n0) n=m+n0; y=x; end </pre>

4	Signal folding	$y(n)=\{x(-n)\}$	<pre>function [y n] = sigfold(x,n) y=fliplr(x); n=-fliplr(n); % fliplr is library function. end >> help fliplr in command window to know in details.</pre>
5	Decomposition to even and odd part	$x_e(n) = \frac{1}{2}[x(n) + x(-n)]$ $x_o(n) = \frac{1}{2}[x(n) - x(-n)]$	<pre>function [xe, xo, m] = evenodd(x,n) m = -fliplr(n); m1 = min([m,n]); m2 = max([m,n]); m = m1:m2; nm = n(1)-m(1); n1 = 1:length(n); x1 = zeros(1,length(m)); x1(n1+nm) = x; x = x1; xe = 0.5*(x + fliplr(x)); xo = 0.5*(x - fliplr(x)); end</pre>

Example 2.7:

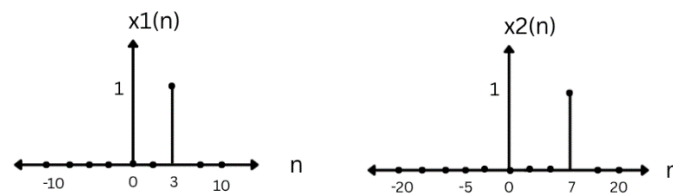


Figure 11 : Delta Signals

Consider two delta signal $x_1(n)$ and $x_2(n)$. Perform the following operations and plot the results:

- Add signal $x_1(n)$ and $x_2(n)$ while multiplying the amplitude of $x_2(n)$ with 3.
- Add signal $x_1(n)$ and $x_2(n)$ while multiplying the amplitude of $x_2(n)$ with -3.
- Consider the added signal from a) as signal $x_3(n)$ and from b) as signal $x_4(n)$. Now multiply signal $x_3(n)$ and $x_4(n)$.
- Perform unit advance shift on signal $x_1(n)$.
- Fold the signal $x_1(n)$.

Performing addition, multiplication, shifting and folding operations using MATLAB code

```
clc; clear all; close all;
%% Signal addition operation
[x1,n1] = delta(3,0,10);
[x2,n2] = delta(7,0,10);
[x3,n3] = sigadd(x1*2,n1,x2*3,n2);
[x4,n4] = sigadd(x1*2,n1,-x2*3,n2);
```

figure

```
stem(n3,x3); axis([-1,11,-0.1,3.4]);
ylabel('x3(n)'); xlabel('Number of Samples (n)');
title('Figure of added signal with different gain'); fontsize(12, "points");
```

figure


```

stem(n4,x4); axis([-1,11,-3.2,2.2]);
ylabel('x4(n)'); xlabel('Number of Samples (n)');
title('Figure of added signal with different gain'); fontsize(12, "points");

%% Signal Multiplication Operation
[x5 n5]=sigmult(x3,n3,x4,n4);

figure
stem(n5,x5); axis([-1,11,-10,5]);
ylabel('x3(n)'); xlabel('Number of Samples (n)');
title('Figure of two multiplied delta signal'); fontsize(12, "points");

%% Signal Shifting Operation
n0=1; %unit shifting
[x6 n6]=sigshift(x1,n1,n0);

figure
subplot(2,1,1);
stem(n1,x1); axis([-1,11,-0.1,1.2]);
ylabel('x1(n)'); xlabel('Number of Samples (n)');
title('Figure of Delta Signal 1'); fontsize(12, "points");

subplot(2,1,2)
stem(n6,x6); axis([-1,12,-0.2,1.2]);
ylabel('x1(n)'); xlabel('Number of Samples (n)');
title('Figure of unit shifted delta signal'); fontsize(12, "points");

%% Signal Folding Operation
[x7 n7] = sigfold(x1,n1);

figure
subplot(2,1,1);
stem(n1,x1); axis([-1,11,-0.1,1.2]);
ylabel('x1(n)'); xlabel('Number of Samples (n)');
title('Figure of Delta Signal 1'); fontsize(12, "points");

subplot(2,1,2);
stem(n7,x7); axis([-12,1,-0.2,1.2]);
ylabel('x1(n)'); xlabel('Number of Samples (n)');
title('Figure of Folded delta signal'); fontsize(12, "points");

```

Output of Example 2.7(a)

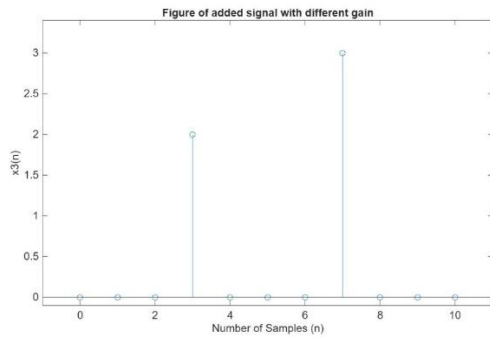


Figure 12 : Addition of different delta signal using MATLAB

Output of Example 2.7(b)

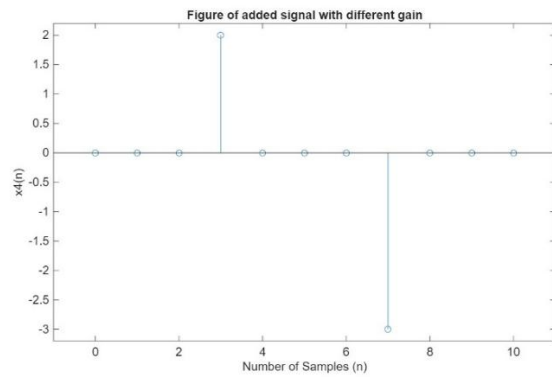


Figure 13 : Multiplication of different delta signal using MATLAB

Output of Example 2.7(c)

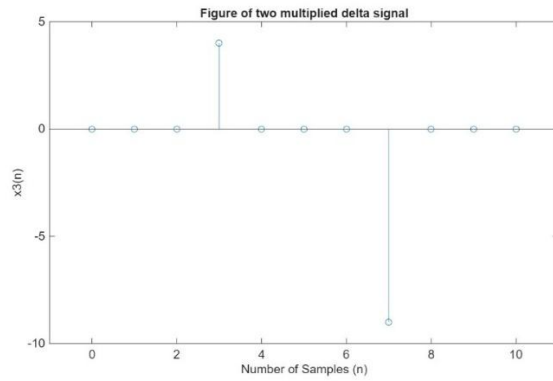


Figure 14: Multiplication of different delta signal using MATLAB

Output of Example 2.7(d)

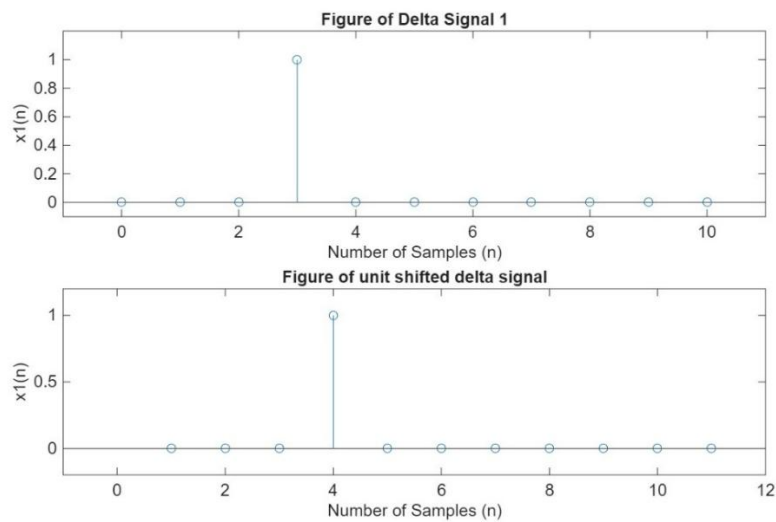


Figure 15 : Unit Delay of a signal using MATLAB

Output of Example 2.7(e)

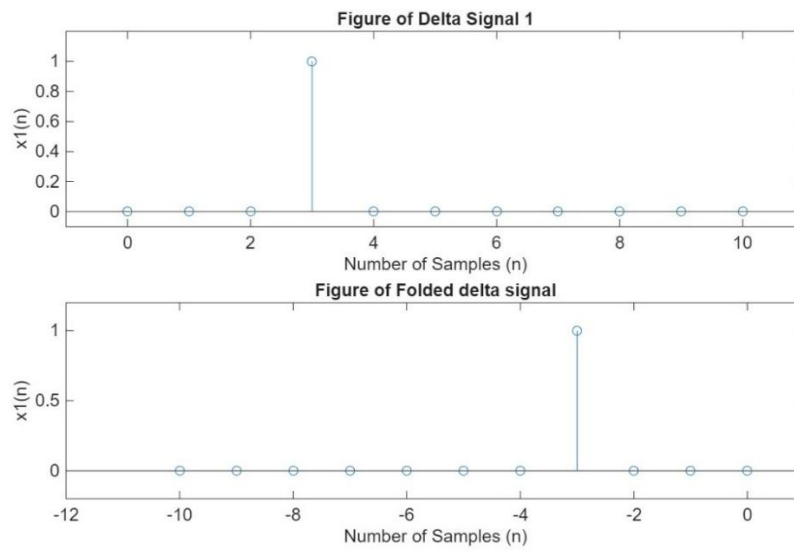


Figure 16 : Flipping a signal using MATLAB

2.4 Exercise

HW2.01:

A simple digital differentiator is given by

$$y(n) = x(n) - x(n - 1)$$

which computes a backward first-order difference of the input sequence. Implement this differentiator on the following sequences, and plot the results. Comment on the appropriateness of this simple differentiator.

(i) $x(n) = 5[u(n) - u(n - 20)]$

(ii) $x(n) = \sin\left(\frac{\pi n}{25}\right)[u(n) - u(n - 100)]$

HW2.02:

Let, $x(n) = \{1, -2, 6, -5, 4, 6, 8, 10\}$ Generate and plot the samples of the following sequence:

$$y(n) = \sum_{k=1}^5 nx(n-k)$$

Hint: Use a for loop and “sigadd” & “sigshift” functions.